Probability Theory

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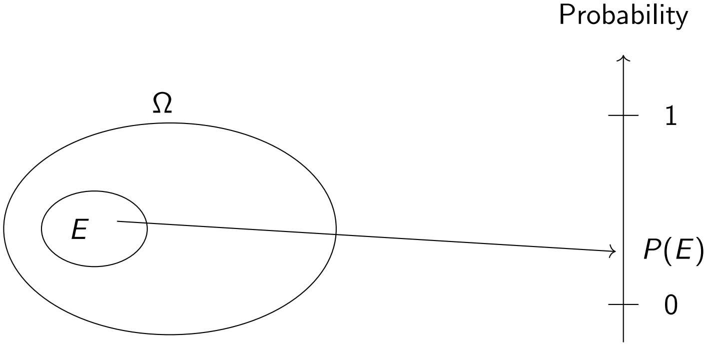
Outline

1. Introduction to Bayes Law
2. Introduction to Conditional Expectations
3. "Normal-Normal" Updating

Bayes Law

Probability Space

An (at most countable) set Ω is a *probability space* if it is endowed with a function (formally, a probability measure) P, which associates, with each subset E of Ω, the probability $P(E)$ that the event E occurs.



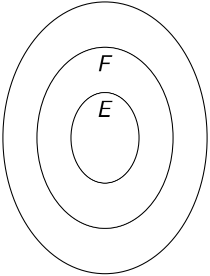
Bayes Law

Probability Measure

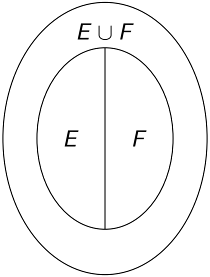
The probability measure P has to satisfy the following properties:

1. $P(\emptyset) = 0$. ($\emptyset$: the event that never happens)
2. $P(\Omega) = 1$. (Ω: the event that is always true)
3. If $E \subseteq F$ then $P(E) \leq P(F)$.
4. If $E \cap F = \emptyset$ then $P(E \cup F) = P(E) + P(F)$.

Ω: $P(\Omega) = 1$



Ω

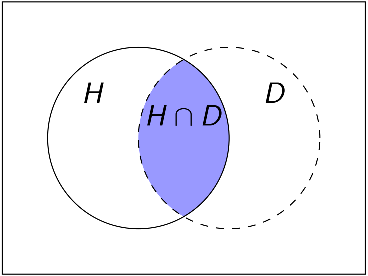


Bayes Law

- D: "Data"
- H: "Hypothesis"

Conditional Probability

$$P(D \mid H) := \frac{P(D \cap H)}{P(H)}, \text{ provided that } P(H) > 0.$$



Bayes Law

Bayes Theorem 1

Let $P(D) > 0$. Then:

$$P(H \mid D) = \frac{P(H)P(D \mid H)}{P(D)}.$$

Conditional probability $P(D \mid H)$: $\underbrace{H}_{\text{Cause}} \longrightarrow \underbrace{D}_{\text{Effect}}$

Bayes Theorem 1 $P(H \mid D)$: $\overbrace{H}^{\text{Hypothesis}} \longleftarrow \overbrace{D}^{\text{Data}}$

Bayes Law

Bayes Theorem 1

Let $P(D) > 0$. Then:

$$P(H \mid D) = \frac{P(H)P(D \mid H)}{P(D)}.$$

Proof.

$\square$

Bayes Law

Bayes Theorem 2

Let $P(D) > 0$. Then:

$$P(H \mid D) = \frac{P(H)P(D \mid H)}{P(H)P(D \mid H) + P(H^c)P(D \mid H^c)}.$$

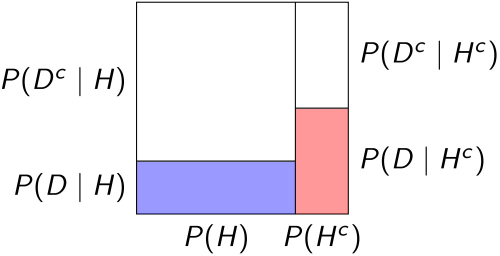


Figure: Bayes Theorem 2

Bayes Law

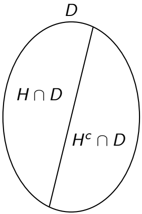
Bayes Theorem 2

Let $P(D) > 0$. Then:

$$P(H \mid D) = \frac{P(H)P(D \mid H)}{P(H)P(D \mid H) + P(H^c)P(D \mid H^c)}.$$

Proof.

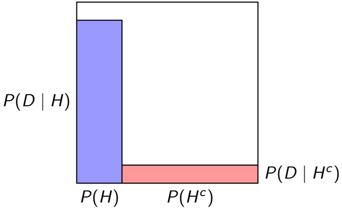
$\square$



Bayes Law

Example

- Suppose that we have a test against a disease that affects 1% of the population.
- If a person has the disease, then the test gets positive with probability 0.95.
- If a person does not have the disease, then the test gets negative with probability 0.9.
- Suppose that a random person tests positive. Now, using the Bayes law, find the posterior probability that he/she has the disease.

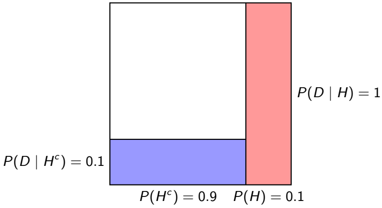


Note: $\frac{19}{217} \approx 0.0876$

Bayes Law

Example (Optional Exercise)

- Suppose that you've been waiting for your suitcase to appear in the carousel at an airport.
- Suppose that, (after having waited for 9 minutes), if your bag was loaded then the probability of finding your bag is 90%.
- Assume that with 90% chance, bags are correctly loaded.
- Find the posterior probability that your bag was not loaded in the plane.



- H: not loaded
- D: Missing your bag
- Then,

$$P(H \mid D) = \frac{0.1 \times 1}{0.1 \times 1 + 0.9 \times 0.1} = \frac{0.1}{0.19} \approx 0.5263.$$

Bayes Law

Market Microstructure Example

Exercise 2.11 (Optional, See also O'Hara, 1998, Section 3.A.1.)

- Suppose that a market maker holds inventory of an asset, which has either a high (H) or low (L) value.
  - The market maker's prior belief that the value of the asset is high is denoted by $p = P(H) \in (0, 1)$.
- There are two types of traders, informed and uninformed, and the market maker meets an informed trader with probability $q \in (0, 1)$.
  - An informed trader always buys when the value is high, and always sells when the value is low.
  - An uninformed trader is equally likely buy or sell.
- Find the posterior probability that the value is high if the first trade is a sale (S), as a function of p and q.

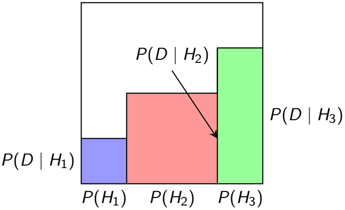
Bayes Law

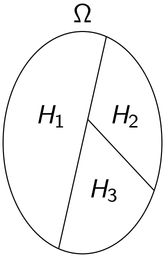
Exercise 2.14 (Optional)

Suppose that a collection of hypotheses $(H_j)_{j=1}^n$ is exhaustive and mutually exclusive (technically, a collection of events $(H_j)_{j=1}^n$ partitions the sample/state space).

Prove the following form of Bayes theorem: for each $i \in \{1, \ldots, n\}$,

$$P(H_i \mid D) = \frac{P(D \mid H_i)P(H_i)}{\sum_{j=1}^n P(D \mid H_j)P(H_j)}.$$

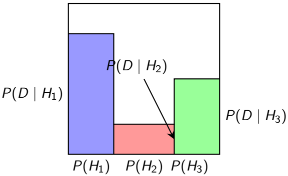




Bayes Law

Exercise 2.15 (Optional)

- Suppose, (in country X), there are two parties: A and B.
- Assume that 30% of people supports party A, 40% party B, and 30% none of them.
- According to a political poll, 80% of party A' supporters, 20% of party B's supporters, and 50% of others support the current cabinet.
- Now, a randomly chosen person was supporting the current cabinet.
- Find the (posterior) probability that that person is supporting party A.



Note: $\frac{24}{47} \approx 0.5106$

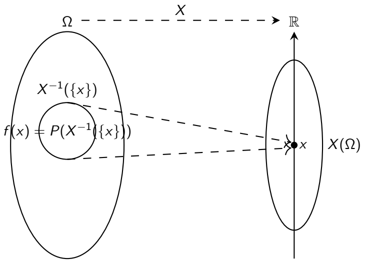
Random Variables

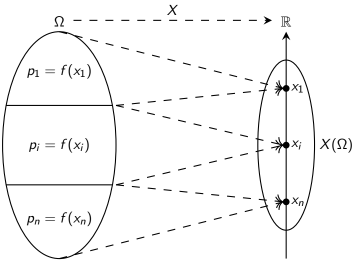
Random Variable

- A *random variable* X (on the probability space Ω) is a function $X : \Omega \to \mathbb{R}$.
-
  - Denoting by $X(\Omega) = \{x_i\}_{i=1}^n$ the set of possible realizations, a function $f : X(\Omega) \to [0,1]$ is a *probability density function* of X if

$$f(x) = P(\{\omega \in \Omega \mid X(\omega) = x\}) \left(= P(X^{-1}(\{x\}))\right) \text{ for all } x \in X(\Omega).$$

  - $f(x_i)$ corresponds to the probability that X takes the value x_i.
  - Write $p_i = f(x_i)$.





Expectations

Expectation

- The *expectation* of X is

$$\mathbb{E}[X] := \sum_{i=1}^{n} p_i x_i.$$

- The expectation can be rewritten as

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} P(\{\omega\}) X(\omega).$$

Ω
$\{\omega \in \Omega \mid X(\omega) = x_1\}$
$\vdots$
$\{\omega \in \Omega \mid X(\omega) = x_n\}$

$\sum_{i=1}^{n} p_i x_i = \sum_{\omega \in \Omega} P(\{\omega\}) X(\omega)$
$p_1 x_1 = \sum_{\omega \in \Omega : X(\omega) = x_1} P(\{\omega\}) X(\omega)$
$\vdots$
$p_n x_n = \sum_{\omega \in \Omega : X(\omega) = x_n} P(\{\omega\}) X(\omega)$

Conditional Expectations

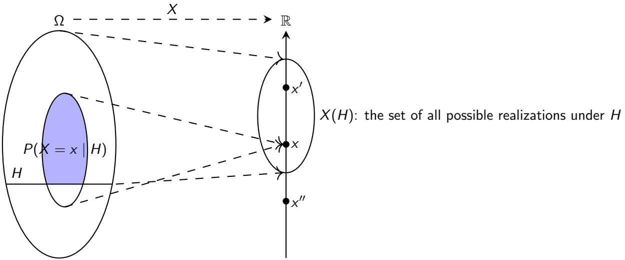
Conditional Expectation

Letting H be an event, the expectation of X conditioned on H is

$$\mathbb{E}[X \mid H] := \sum_{x \in X(H)} P(X = x \mid H)x.$$

This can be rewritten as

$$\mathbb{E}[X \mid H] = \sum_{\omega \in H} P(\{\omega\} \mid H)X(\omega).$$

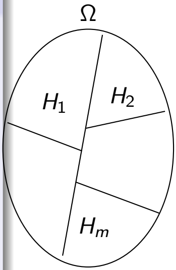


Law of Iterated Expectations

Theorem 2.4: Law of Iterated Expectations

Suppose that a collection of events $(H_j)_{j=1}^m$ is exhaustive and mutually exclusive (i.e., $(H_j)_{j=1}^m$ partitions the sample space). Suppose $P(H_j) > 0$ for all $j \in \{1, \ldots, m\}$. Let X be a random variable. Then,

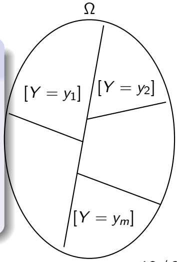
$$\mathbb{E}[X] = \sum_{j=1}^{m} \underbrace{\mathbb{E}[X \mid H_j]}_{\text{function of } H_j} P(H_j) \, (= \mathbb{E}[\mathbb{E}[X \mid H_j]]) \, .$$



Theorem 2.5: Law of Iterated Expectations

Let X and Y be a random variable. Then,

$$\mathbb{E}[X] = \mathbb{E}[\, \underbrace{\mathbb{E}[X \mid Y]}_{\text{function of } Y} \,].$$



Law of Iterated Expectations

$$\mathbb{E}[X] = \sum_{j=1}^{m} \mathbb{E}[X \mid H_j] P(H_j) \, (= \mathbb{E}[\mathbb{E}[X \mid H_j]]) \, .$$

$\mathbb{E}[X] = \sum_{\omega \in \Omega} P(\{\omega\}) X(\omega)$

$p_1 x_1$	$\sum_{\omega \in \Omega : X(\omega) = x_1} P(\{\omega\}) X(\omega)$
$p_2 x_2$	$\sum_{\omega \in \Omega : X(\omega) = x_2} P(\{\omega\}) X(\omega)$
	$\vdots$
$p_n x_n$	$\sum_{\omega \in \Omega : X(\omega) = x_n} P(\{\omega\}) X(\omega)$

$\sum_{j=1}^{m} \mathbb{E}[X \mid H_j] P(H_j) = \sum_{\omega \in \Omega} P(\{\omega\}) X(\omega)$

$\mathbb{E}[X \mid H_1] P(H_1) = \sum_{\omega \in H_1} P(\{\omega\}) X(\omega)$
$\mathbb{E}[X \mid H_2] P(H_2) = \sum_{\omega \in H_2} P(\{\omega\}) X(\omega)$
$\vdots$
$\mathbb{E}[X \mid H_m] P(H_m) = \sum_{\omega \in H_m} P(\{\omega\}) X(\omega)$

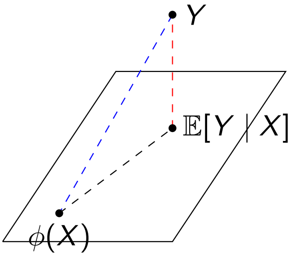
$$\begin{aligned} \mathbb{E}[X \mid H_j] P(H_j) &= \sum_{\omega \in H_j} P(\{\omega\} \mid H_j) X(\omega) P(H_j) \\ &= \sum_{\omega \in H_j} P(\{\omega\}) X(\omega). \end{aligned}$$

Conditional Expectation as the Best Predictor

Best Predictor (Optional Exercise)

Show that, for any function ϕ of X,

$$\mathbb{E}[(Y - \phi(X))^2] \geq \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2].$$



Conditional Expectation as the Best Predictor

Proof.

First, we have

$$
\begin{aligned}
&\mathbb{E}[(Y-\phi(X))^2] \\
=&\mathbb{E}[(Y-\mathbb{E}[Y \mid X]+\mathbb{E}[Y \mid X]-\phi(X))^2] \\
=&\mathbb{E}[(Y-\mathbb{E}[Y \mid X])^2+2(Y-\mathbb{E}[Y \mid X])(\mathbb{E}[Y \mid X]-\phi(X))+\underbrace{(\mathbb{E}[Y \mid X]-\phi(X))^2}_{\geq 0}].
\end{aligned}
$$

Next, by the law of iterated expectations,

$$
\begin{aligned}
\mathbb{E}[(Y-\mathbb{E}[Y \mid X])(\mathbb{E}[Y \mid X]-\phi(X))] &= \mathbb{E}[\mathbb{E}[(Y-\mathbb{E}[Y \mid X])(\mathbb{E}[Y \mid X]-\phi(X)) \mid X]] \\
&= \mathbb{E}[\underbrace{(\mathbb{E}[Y \mid X]-\mathbb{E}[Y \mid X])}_{=0}(\mathbb{E}[Y \mid X]-\phi(X))] \\
&= 0.
\end{aligned}
$$

Hence, we obtain

$$
\begin{aligned}
\mathbb{E}[(Y-\phi(X))^2] &= \mathbb{E}[(Y-\mathbb{E}[Y \mid X])^2]+\mathbb{E}[(\mathbb{E}[Y \mid X]-\phi(X))^2] \\
&\geq \mathbb{E}[(Y-\mathbb{E}[Y \mid X])^2].
\end{aligned}
$$

Bayes Law: "Normal-Normal" Case

"Normal-Normal" Updating

- Suppose a random variable $X \sim N\left(\mu_x, \sigma_x^2\right)$ is observed with error

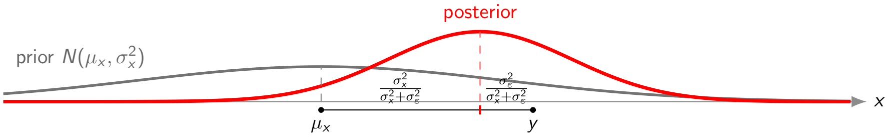
$$Y = X + \varepsilon,$$

where $\varepsilon \sim N\left(0, \sigma_\varepsilon^2\right)$ is a random variable independent of X.

- The posterior for X given $Y = y$ follows a normal distribution:

$$X \mid Y = y \sim N\left(\frac{\sigma_\varepsilon^2 \mu_x + \sigma_x^2 y}{\sigma_x^2 + \sigma_\varepsilon^2}, \frac{\sigma_x^2 \sigma_\varepsilon^2}{\sigma_x^2 + \sigma_\varepsilon^2}\right).$$

- Mean: $\frac{\sigma_\varepsilon^2}{\sigma_x^2+\sigma_\varepsilon^2}\mu_x + \left(1 - \frac{\sigma_\varepsilon^2}{\sigma_x^2+\sigma_\varepsilon^2}\right) y = \mu_x + \frac{\sigma_x^2}{\sigma_x^2+\sigma_\varepsilon^2}(y - \mu_x)$;
- Variance: $\frac{\sigma_\varepsilon^2}{\sigma_x^2+\sigma_\varepsilon^2}\sigma_x^2 = \sigma_x^2 - \frac{\sigma_x^2}{\sigma_x^2+\sigma_\varepsilon^2}\sigma_x^2 < \sigma_x^2$.



Bayes Law: "Normal-Normal" Case

Derivation

- By the Bayes law, the conditional probability density function is

$$f(\underbrace{x}_{H} \mid \underbrace{Y=y}_{D}) = \frac{\overbrace{\pi(x)L(y \mid x)}^{=P(X=x,Y=y)}}{\int_{-\infty}^{\infty} \underbrace{\pi(\tilde{x})L(y \mid \tilde{x})}_{=P(X=\tilde{x},Y=y)}\, d\tilde{x}},$$

where

$$\pi(x) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{(x-\mu_x)^2}{2\sigma_x^2}\right) \text{ and}$$

$$L(y \mid x) = \frac{1}{\sqrt{2\pi}\sigma_\varepsilon} \exp\left(-\frac{(y-x)^2}{2\sigma_\varepsilon^2}\right).$$

Bayes Law: "Normal-Normal" Case

Derivation

- Now,

$$\begin{aligned} f(x \mid Y = y) &\propto \pi(x) L(y \mid x) \\ &\propto \exp\left(-\frac{(x-\mu_x)^2}{2\sigma_x^2}\right) \exp\left(-\frac{(y-x)^2}{2\sigma_\varepsilon^2}\right) \\ &= \exp\left(-\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-x)^2}{2\sigma_\varepsilon^2}\right). \end{aligned}$$

Bayes Law: "Normal-Normal" Case

Derivation

- Thus, compute:

$$
\begin{aligned}
& -\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-x)^2}{2\sigma_\varepsilon^2} \\
&= -\frac{1}{2\sigma_x^2\sigma_\varepsilon^2}\left(\sigma_\varepsilon^2(x-\mu_x)^2 + \sigma_x^2(y-x)^2\right) \\
&= -\frac{1}{2\sigma_x^2\sigma_\varepsilon^2}\left((\sigma_\varepsilon^2+\sigma_x^2)x^2 - 2(\sigma_\varepsilon^2\mu_x + \sigma_x^2 y)x + (\sigma_\varepsilon^2\mu_x^2 + \sigma_x^2 y^2)\right) \\
&= -\frac{\sigma_\varepsilon^2+\sigma_x^2}{2\sigma_x^2\sigma_\varepsilon^2}\left(x - \frac{\sigma_\varepsilon^2\mu_x + \sigma_x^2 y}{\sigma_\varepsilon^2+\sigma_x^2}\right)^2 + (\text{something}).
\end{aligned}
$$

- Then,

$$
\begin{aligned}
f(x \mid Y=y) &\propto \exp\left(-\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-x)^2}{2\sigma_\varepsilon^2}\right) \\
&\propto \exp\left(-\frac{1}{2\frac{\sigma_x^2\sigma_\varepsilon^2}{\sigma_\varepsilon^2+\sigma_x^2}}\left(x - \frac{\sigma_\varepsilon^2\mu_x + \sigma_x^2 y}{\sigma_\varepsilon^2+\sigma_x^2}\right)^2\right)
\end{aligned}
$$

Bayes Law: "Normal-Normal" Case

Derivation

- This means:

$$X \mid Y = y \sim N\left(\frac{\sigma_\varepsilon^2 \mu_x + \sigma_x^2 y}{\sigma_\varepsilon^2 + \sigma_x^2}, \frac{\sigma_x^2 \sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_x^2}\right)$$

and indeed

$$f(x \mid Y = y) = \frac{1}{\sqrt{2\pi \frac{\sigma_x^2 \sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_x^2}}} \exp\left(-\frac{1}{2\frac{\sigma_x^2 \sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_x^2}}\left(x - \frac{\sigma_\varepsilon^2 \mu_x + \sigma_x^2 y}{\sigma_\varepsilon^2 + \sigma_x^2}\right)^2\right).$$

Bayes Law: "Normal-Normal" Case

- It's convenient to recast the "normal-normal" updating in terms of precision ($= 1/\text{variance}$):

$$\tau_x = \frac{1}{\sigma_x^2} \text{ and } \tau_\varepsilon = \frac{1}{\sigma_\varepsilon^2}.$$

"Normal-Normal" Updating

- The posterior for $X \sim N(\mu_x, \tau_x^{-1})$ given $Y = y$, where $Y = X + \varepsilon$ with $\varepsilon \sim N(0, \tau_\varepsilon^{-1})$ and $\varepsilon \perp X$, follows a normal distribution:

$$X \mid Y = y \sim N\left(\frac{\tau_x \mu_x + \tau_\varepsilon y}{\tau}, \tau^{-1}\right),$$

where

$$\tau = \tau_x + \tau_\varepsilon.$$

Bayes Law: "Normal-Normal" Case

"Normal-Normal" Updating, Multiple Observations

- Suppose $X \sim N(\mu_x, \tau_x^{-1})$ is observed with independent error

$$y_j = X + \varepsilon_j \text{ for each } j \in \{1, \ldots, n\},$$

where $\varepsilon_j \sim N(0, \tau_j^{-1})$.

- The posterior for X given observations $(Y_j = y_j)_{j=1}^n$ follows a normal distribution:

$$x \mid (Y_j = y_j)_{j=1}^n \sim N\left(\overbrace{\frac{\tau_x}{\tau}\mu_x + \sum_{j=1}^{n} \frac{\tau_j}{\tau} y_j}^{\substack{\text{weighted average of} \\ \mu_x \text{ and } (y_j)_{j=1}^n}}, \tau^{-1} \right),$$

where

$$\tau = \tau_x + \sum_{j=1}^{n} \tau_j.$$

Bayes Law: "Normal-Normal" Case

Derivation (Optional)

- By the Bayes Law,

$$
\begin{aligned}
f\left(x \mid (Y_j = y_j)_{j=1}^n\right) &\propto \pi(x) \prod_{j=1}^n L(y_j \mid x) \\
&\propto \exp\left(-\frac{\tau_x}{2}(x-\mu_x)^2\right) \prod_{j=1}^n \exp\left(-\frac{\tau_j}{2}(y_j - x)^2\right) \\
&= \exp\left(-\frac{1}{2}\left(\tau_x(x-\mu_x)^2 + \sum_{j=1}^n \tau_j (x - y_j)^2\right)\right).
\end{aligned}
$$

Bayes Law: "Normal-Normal" Case

Derivation (Optional)

- Then,

$$
\begin{aligned}
&\tau_x(x-\mu_x)^2 + \sum_{j=1}^{n} \tau_j (x - y_j)^2 \\
&= \left(\tau_x + \sum_{j=1}^{n} \tau_j\right) x^2 - 2\left(\tau_x \mu_x + \sum_{j=1}^{n} \tau_j y_j\right) x + (\text{something}) \\
&= \left(\tau_x + \sum_{j=1}^{n} \tau_j\right) \left(x - \frac{\tau_x \mu_x + \sum_{j=1}^{n} \tau_j y_j}{\tau_x + \sum_{j=1}^{n} \tau_j}\right)^2 + (\text{something}).
\end{aligned}
$$

Bayes Law: "Normal-Normal" Case

Derivation (Optional)

- Letting
$$\tau = \tau_x + \sum_{j=1}^{n} \tau_j,$$
the posterior for $x \mid (Y_j = y_j)_{j=1}^{n}$ follows a normal distribution with mean
$$\frac{\tau_x \mu_x + \sum_{j=1}^{n} \tau_j y_j}{\tau_x + \sum_{j=1}^{n} \tau_j} = \frac{\tau_x}{\tau} \mu_x + \sum_{j=1}^{n} \frac{\tau_j}{\tau} y_j$$
and variance
$$\tau^{-1}.$$

Bayes Law: "Normal-Normal" Case

List of Applications

- Microeconomic Theory
  - Herding, Information aggregation, Career concern, Global game, Team theory, Network, ...
- Macroeconomic Theory and Finance
  - Market microstructure (e.g., Kyle), Rational expectation (e.g., Grossman-Stiglitz), Portfolio choice (in the CARA-Normal setting), Social value of information/Transparency (e.g., Morris-Shin), ...
- Econometrics
  - Time series (Kalman filtering), ...